DSFDcd: Joint Distribution Sampling and Feature Decoupling Deep Network for Remote Sensing Change Detection

Bin Wang, Xiaohu Jiang, Pinle Qin, *Member, IEEE*, and Jianchao Zeng

***Abstract*— Remote sensing change detection (RSCD) aims to identify the regions of interest that have changed between dual-temporal images. However, most deep models predict CD results by extracting multiscale hybrid features, which could easily lead to ambiguous semantic boundaries; in addition, the existing feature acquisition tends to lack consideration of capturing their diversity, usually causing poor model generalization. Thus, this article decomposes the mixed features into change and invariant features jointly with stochastic distribution sampling and convolution, thus accomplishing robust RSCD based on decoupled representations. In the training stage, the posterior distribution of the uncoupled features is first learned through label calibration to train the prior distribution generator; then, robust feature decoupling is implemented combining the convolutional feature separator with reparameterized sampling over the decoupled posteriori distribution, and further aggregating the decoupled features through prototype learning; finally, the exceed-expectation (EE) loss regularizer is proposed to push or pull these positive and negative sample features to a more distant end, thereby increasing the interclass distance by boosting the predicted expectation. In the testing stage, the robust RSCD based on decoupled representation is accomplished through the feature separator, decoupled prior distribution random sampling, and the CD head without posterior distribution support. Experiments prove that DSFDcd has achieved remarkable results in terms of qualitative and quantitative metrics. Our codes will be available at https://github.com/iceking111/DSFDcd**

***Index Terms*— Change detection, distribution sampling, exceed-expectation (EE) loss, feature decoupling.**

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I. Introduction

REMOTE sensing change detection (RSCD) aims to compare the differences between images of the same area at different times and mark the changed areas. It has been widely applied in fields such as urban expansion analysis, disaster assessment, military strikes, and vegetation cover detection [1], [2], [3], [4].

Traditional CD methods can be broadly classified into three categories [5], [6], [7]: methods based on image arithmetic, image transformation, and postclassification, and these methods tend to be less generalizable and robust. The development of machine learning has improved the CD accuracy to some extent. However, when faced with different scenarios, the selection of machine learning methods and their generalization pose challenges for practical applications. Owing to powerful feature extraction and nonlinear representation capabilities [8], [9], [10], deep learning methods have been extensively applied to CD tasks and have demonstrated excellent performance.

Representative deep network frameworks for RSCD include convolutional neural networks (CNNs), recurrent neural networks, Transformer, Mamba, and their hybrid models [11], [12], [13]. Many researchers have accomplished CD tasks based on the fully convolutional network and U-Net architectures. Furthermore, the another mainstream about CNN-based CD network is the Siamese network. It extracts deep features from bi-temporal images through a two-stream architecture, and then fuses and enhances relevant features to obtain CD results. To explore the temporal correlations within the feature tensors of bi-temporal images, some researchers have proposed CD models that integrate a long short-term memory net. Notably, to enable neural networks to accurately focus on important features and suppress unimportant ones, some research efforts have utilized the attention mechanism to identify the areas of change of interest. Such methods mainly achieve feature reweighting through various means (such as channel, spatial, and correlation-based methods) to highlight the effective information, ultimately enhancing the CD performance. However, convolution-based deep networks often suffer from problems such as computational burden or poor robustness. For this reason, some researchers have constructed brain-inspired frameworks based on spiking neural network with low power consumption and strong generalization ability for relevant RS scenarios, such as classification [14], [15], [16], [17], optical flow estimation [18], and so on.

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To identify and correlate changed areas at different times and locations, some studies establish long-range dependency relationships using Transformer [19], [20], [21]. This approach can parallelly learn useful information in different representational subspaces through the multihead attention, without being restricted by their positions. However, Transformer-based models typically contain a large number of parameters. As a result, they often consume a vast amount of computational resources and training time [22], [23]. In addition, Hong et al. [24] developed a Transformer-based general RS fundamental model named SpectralGPT with a large amount of spectral data, which can be generalized with downstream RS tasks, such as landmark classification, and CD. It is well known that foundation models require big data to sustain them, and flexibly more and more high-quality RS data are being produced and shared by researchers [25].

In addition, Mamba has recently been applied in the field of RSCD. This model incorporates an omni-selective scanning module, which can selectively scan images in multiple directions [26], [27], [28]. This allows it to capture large-scale spatial features from different orientations. The application of the Mamba model in RSCD demonstrates its powerful capabilities in modeling spatio-temporal relationships and efficiently capturing global context. However, due to multiple temporal scans involved, Mamba-based RSCD models also consume huge computational resources.

These CD models based on mainstream deep learning frameworks like CNN, Transformer, and Mamba are generally accomplished through hybrid feature extraction and fusion, followed by pixel-level classification. However, hybrid features may impede the model's ability to learn the essence of both changing and unchanging features, and it is susceptible to cause semantic boundary ambiguity for pixel classification which in turn affects the model performance as shown in Fig. 1. Consequently, there is a tendency toward increasing model complexity and parameter number for more higher indicators. In addition, improving the model's generalization has always been a research spotlight in the RSCD field.

To address the above-mentioned issues, this article proposes an RSCD framework based on feature decoupling. The aim is to decompose the bi-temporal features into change-related and unchanged features, thus enabling change interpretation. As shown in Fig. 2, the posterior probability distributions of disentangled features are first learned from the difference features and labels. These distributions are then utilized to guide the prior probability feature generator. Second, the changed and unchanged features are separately obtained by jointly using the feature separator and posterior distribution reparameterization sampling. In addition, prototype learning is employed to make the decoupled features more compact. Finally, the binary cross entropy (BCE) and hyper-expected push-pull hybrid loss function is used to increase the interclass distance between positive and negative samples, making it easier to distinguish between changed and unchanged pixels.

Compared with existing RSCD methods, our model exhibits two main properties: first, it effectively separates hybrid features and reveals the potential mechanism of change generation more clearly through cross-domain disentanglement

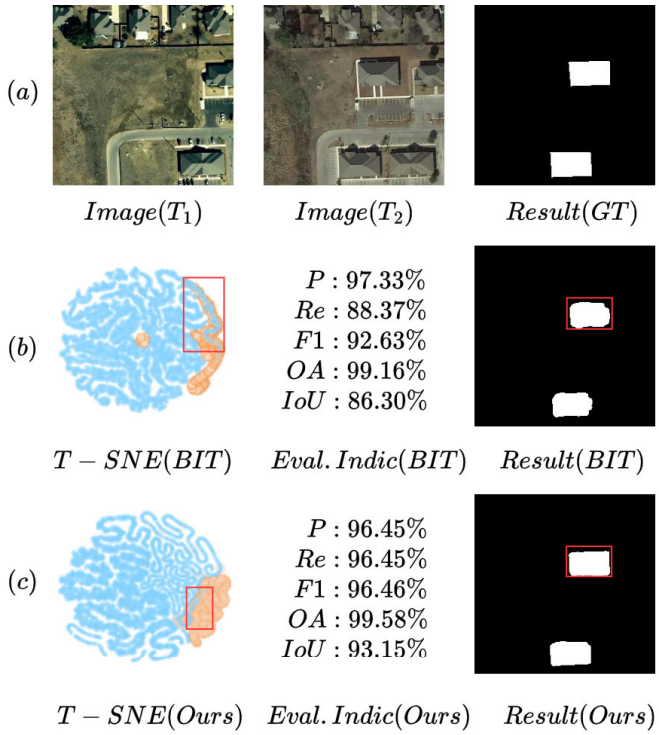


Fig. 1. Visualization comparison of changing and unchanging pixel classification. (a) Bi-temporal RS images and GT. (b) Example of the classical BIT: T-SNE visualization, objective metrics, and prediction results. (c) Example of our model: T-SNE visualization, objective metrics, and prediction results. Obviously, the semantic boundaries are clearer with respect to the CD results generated by the proposed DSFDcd based on decoupled representations and diverse feature extraction.

representation; second, it generates diversified features that facilitate the model to understand complex RS scenes from multiple perspectives, and the ideology can be spread to other vision tasks. The contributions of this article are mainly as follows.

1) A novel RSCD framework based on decoupled representation and diverse feature extraction is proposed to project hybrid features into the variable and invariant feature space. Meanwhile, the learned latent space for disentangled features could be utilized to enhance the feature diversity by stochastic sampling.
2) Prototype learning is introduced to constrain the diverse decoupled features, while the disentangled feature separator and its prior distribution generator are trained synchronously to make the RSCD decoupling space more compact.
3) The exceeding expectations push-pull loss regularization is proposed, which together with cross entropy can effectively increase the distance between positive and negative sample classes to directly benefit pixel classification.

The remaining part of this article is organized as follows. Section II briefly describes related work. Section III mainly depicts the model framework and details. Section IV introduces the experimental setup. Section V presents the experimental results. Section VI is the conclusion.

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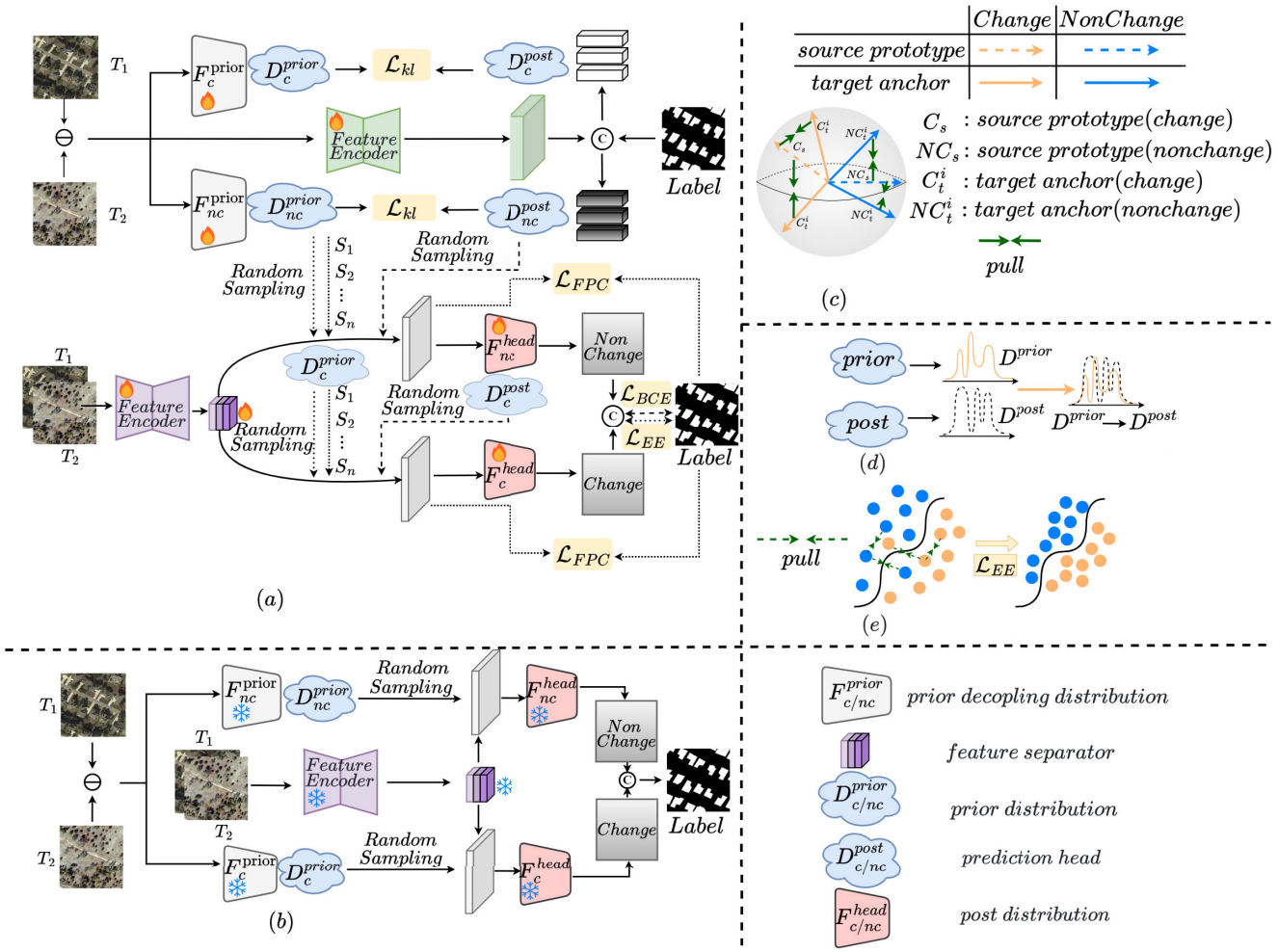


Fig. 2. Deep network for RSCD using joint distribution sampling and feature decoupling. During the training phase, the prior feature generator, feature extractor, and feature decoupler are mainly trained through posterior features and prototype learning. In the testing phase, the features of bi-temporal images are decomposed into changing and invariant features, and the final CD result is generated based on the trained modules. (a) Training stage. (b) Testing stage. (c) FPC. (d) Minimizing KL($D^{\text{prior}}||D^{\text{post}}$). (e) EE loss.

II. Related Work

A. Feature Decoupling

Feature decoupling representation aims to separate mixed features into different and relatively independent subspaces or representation forms through specific methods and model structures. In this way, each subspace can represent a certain specific feature or factor of the data. Feature decoupling representation is more effective in learning the essential feature, making it easier to understand which specific features the model bases its decisions on [29], [30], and [31]. RSCD usually separates features of different nature from the image, such as content and style [32], [33], semantic and spatial [34], [35], and so on, so that the model can understand and process the change information more clearly. Dai et al. [36] disentangled the deep features of dual temporal images into content and style features, and directly calculated the change probability map based on content similarity. Cheng et al. [37] embedded each input image into two different subspaces: a shared content space and a private style space. The style space aimed to alleviate style differences caused by imaging conditions, while the content space reflected semantic features.

B. Prototype Learning

Prototype learning focuses on the feature representation of an entire category. A prototype can be a vector, a set of features, or a model, and so on [38], [39], [40]. In the classification task, an input sample can be mapped to the same space as that of the prototype. Then, a similarity metric (such as Euclidean distance or Cosine similarity) is defined to measure the proximity between the data point and the prototype, and a final judgment is made based on this measurement. Since the prototype is a comprehensive representation of a class of samples, the relevant downstream tasks learned based on the prototype are not disturbed by individual noisy samples, thus improving the model's robustness against data noise [41], [42]. In the RSCD field, researchers could learn category prototype feature vectors for various types of geographical objects (e.g., buildings, vegetation, and water bodies) from the training samples; they could also construct area categories or change states (e.g., "unchanged" and "vegetation to building") of the prototypes. Wu et al. [43] introduced an unbalanced contrastive learning method based on multiple prototypes, which allowed for a more flexible selection of the sample contrasts, adapting to variations among samples of the same class. Zhao et al. [44]

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designed prototype-based contrastive learning, which enables unreliable pixels to learn from prototypes calculated from reliable pixels. Zhang et al. [45] proposed an unpaired image prototype contrast module, which enriched the diversity of the change samples for training by combining the prototypes in unpaired images at the feature level.

C. Distribution Sampling Learning

Distribution sampling learning generally refers to extracting features from a given or learned data distribution for model training [46], [47]. Through a reasonable way, sample features that can represent the overall data distribution can be obtained, enabling the model to learn the data's internal patterns. It is worth mentioning the reparameterization sampling, which represents the sampling process as a combination of a deterministic function and a noise variable. This method allows for sampling from complex probability distributions while also enabling effective gradient calculation [48], [49]. During the training stage, features can be sampled from the learned data distribution for model tuning. This significantly reduces the dependence on feature extraction operators such as convolution and self-attention, and also decreases the demand for computing resources. In addition, the randomness generated by the sampling process can produce more diverse features, thereby effectively improving model generalization and stability [50], [51]. With the rapid development of denoising diffusion probabilistic model (DDPM), distribution sampling learning is also gradually involved in the domain of RSCD. Unlike discriminative models that aimed to classify each pixel into changed or unchanged categories, Wen et al. [52] proposed GCD-DDPM, which could employ an end-to-end architecture to generate a CD map directly by exploiting the DDPM model. To improve model robustness, Li et al. [53] proposed a diffusion-guided spatial–frequency domain information interaction RSCD method, which generated part of the pseudo images by using a conditional DDPM. Notably, reparameterized sampling has been innovatively adopted in the DDPM in order to backpropagate the gradient when restoring the generated image (original image or change mask) from noisy data.

III. Method

A. Motivation and Overview

For complex RS scenes, diversified features assist the model in parsing scene information from multiple perspectives, and rich diversity could also make the model effectively capture multimodal distributions (e.g., different lighting conditions and feature types) in RS data, thus reducing the risk of model overfitting. However, how to effectively enhance the diversity of feature extraction is still a major challenge in the CD field. Inspired by the above-mentioned analyses and the literature [54], we would intend to learn a latent decoupling space, which can be sampled to generate diversified disentangled features for RSCD, and then finally be uniformly inputted into the CD head to generate the decision results.

Fig. 2 illustrates the overall flow of the DSFDcd framework for RSCD, which consists of two main phases: training and testing. The training phase aims to decompose the dual-phase image difference features into change and invariant parts. The modules to be learned include the U-Net feature extractor F_{FE}, the feature separator F_{FS}, the change and invariant feature distribution encoder (i.e., $F_{\mathrm{c}}^{\mathrm{prior}}$ and $F_{\mathrm{nc}}^{\mathrm{prior}}$), and the CD prediction head (i.e., $F_{\mathrm{c}}^{\mathrm{head}}$ and $F_{\mathrm{nc}}^{\mathrm{head}}$); in the testing phase, the main purpose is to freeze the weights of the above mentioned modules, and to predict the change results without the support of the posterior features.

B. Decoupling Feature Distributions (DFDs)

Feature extraction based on convolution preserves spatial location information effectively, while features obtained through distribution sampling can introduce randomness, making the model more robust against data variations. To this end, this article combines their advantages to complete feature decoupling.

First, the difference feature X of the dual-temporal images $T_1, T_2 \in \mathbb{R}^{H\times W\times 3}$ is used independently. Two separate encoders are employed to generate decoupled feature prior distributions, namely, the change prior distribution $\mathbb{D}_{\mathrm{c}}^{\mathrm{prior}}(X)$ and the nonchange prior distribution $\mathbb{D}_{\mathrm{nc}}^{\mathrm{prior}}(X)$. Moreover, both distributions are set as multivariate normal distributions with diagonal covariance matrices, denoted as $\mathcal{N}(\mu_{\mathrm{c}}^{\mathrm{prior}}, \sigma_{\mathrm{c}}^{\mathrm{prior}})$ and $\mathcal{N}(\mu_{\mathrm{nc}}^{\mathrm{prior}}, \sigma_{\mathrm{nc}}^{\mathrm{prior}})$, respectively. The mean and variance of the decoupled feature prior distributions is acquired as follows:

$$\mu_{\mathrm{c}}^{\mathrm{prior}}, \sigma_{\mathrm{c}}^{\mathrm{prior}} = F_{\mathrm{c}}^{\mathrm{prior}}(X) \tag{1}$$

$$\mu_{\mathrm{nc}}^{\mathrm{prior}}, \sigma_{\mathrm{nc}}^{\mathrm{prior}} = F_{\mathrm{nc}}^{\mathrm{prior}}(X) \tag{2}$$

where $F_{\mathrm{c}}^{\mathrm{prior}}$ and $F_{\mathrm{nc}}^{\mathrm{prior}}$ are two learnable prior decoupling distribution generators, as shown in the upper left part of Fig. 2(a).

In order to generate posterior features, a U-Net encoder E_{u1} is used to encode the bi-temporal difference features X^*. Subsequently, the posterior decoupled features are selected through label (i.e., GT). Furthermore, the decoupled posterior distributions are obtained through learning. That is, $\mathbb{D}_{\mathrm{c}}^{\mathrm{post}}(X^*, \mathrm{label}) = \mathcal{N}(\mu_{\mathrm{c}}^{\mathrm{post}}, \sigma_{\mathrm{c}}^{\mathrm{post}})$ and $\mathbb{D}_{\mathrm{nc}}^{\mathrm{post}}(X^*, (1-\mathrm{label})) = \mathcal{N}(\mu_{\mathrm{nc}}^{\mathrm{post}}, \sigma_{\mathrm{nc}}^{\mathrm{post}})$. The mean and variance of the decoupled posterior distributions is obtained as follows:

$$\mu_{\mathrm{c}}^{\mathrm{post}}, \sigma_{\mathrm{c}}^{\mathrm{post}} = F_{\mathrm{c}}^{\mathrm{post}}\left(X^*, \mathrm{label}\right) \tag{3}$$

$$\mu_{\mathrm{nc}}^{\mathrm{post}}, \sigma_{\mathrm{nc}}^{\mathrm{post}} = F_{\mathrm{nc}}^{\mathrm{post}}\left(X^*, (1-\mathrm{label})\right) \tag{4}$$

where $F_{\mathrm{c}}^{\mathrm{post}}$ and $F_{\mathrm{nc}}^{\mathrm{post}}$ are two learnable posterior decoupling distribution generators, as shown in the upper left part of Fig. 2(a). To train the prior distribution generator via the posterior features, the Kullback–Leibler (KL) divergence loss is employed to align the two feature distributions within the decoupled space, respectively,

$$\mathcal{L}_{\mathrm{kl}} = \mathrm{KL}\left(\mathbb{D}_{\mathrm{c}}^{\mathrm{prior}}(X), \mathbb{D}_{\mathrm{c}}^{\mathrm{post}}\left(X^*, \mathrm{label}\right)\right) + \mathrm{KL}\left(\mathbb{D}_{\mathrm{nc}}^{\mathrm{prior}}(X), \mathbb{D}_{\mathrm{nc}}^{\mathrm{post}}\left(X^*, (1-\mathrm{label})\right)\right). \tag{5}$$

The next concentration will focus on CD disentanglement target generation. Given the posterior distribution $\mathbb{D}_{\mathrm{c}}^{\mathrm{post}}$ in the training phase, a latent code $z_{\mathrm{random}}^{\mathrm{post}} \in \mathbb{R}^{D\times 1\times 1}$ is randomly sampled, which is then further scaled to $\hat{z}_{\mathrm{random}}^{\mathrm{post}} \in \mathbb{R}^{D\times H\times W}$, where D represents the dimension of all latent

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codes. Subsequently, the scaled latent code is concatenated with the decoupled features (s_c and s_{nc}) generated by the feature separator, and then fed into the prediction head F_θ^{head} to facilitate feature decomposition and generate the corresponding decoupled results

$$P_c^{\text{post}} = F_c^{\text{head}}\left(\hat{z}_{\text{random}}^{\text{post}}, s_c\right) \tag{6}$$

$$P_{nc}^{\text{post}} = F_{nc}^{\text{head}}\left(\hat{z}_{\text{random}}^{\text{post}}, s_{nc}\right). \tag{7}$$

Consequently, the final change mask can be obtained by combining the change probability map predicted from the posterior distribution, i.e., $P^{\text{post}} = P_c^{\text{post}} \copyright P_{nc}^{\text{post}}$, and performing a pixel-wise argmax operation along its channel dimension. For training the prior distribution generator to assist in feature decoupling and complete the CD task, features would also be sampled from the prior distribution to obtain the change mask p^{prior}. Certainly, stochastic sampling will only occur with a well-trained prior distribution in the testing phase. It is worth noting that in the training phase, we both utilize the reparameterization technique to sample the latent coding from the decoupled feature distribution in order to back-propagate the gradient for model training. In the testing phase, we instead perform regular random sampling directly from the well-trained prior distributions only.

In terms of the CD loss function, network parameters are optimized by minimizing the BCE

$$\mathcal{L}_{\text{BCE}}^{\text{prior}} = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} l_{\text{bce}}\left(P_{hw}^{\text{prior}}, Y_{hw}\right) \tag{8}$$

$$\mathcal{L}_{\text{BCE}}^{\text{post}} = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} l_{\text{bce}}\left(P_{hw}^{\text{post}}, Y_{hw}\right). \tag{9}$$

C. Feature Prototype Constraint (FPC)

Prototype learning can utilize typical data features to train network parameters. In this article, the decoupled space is made more compact by constraining the distance from decoupled feature to their prototype, as shown in Fig. 3.

First, we stochastically sample K decoupled latent codes $\{\hat{z}_c^{\text{prior}}, \hat{z}_{nc}^{\text{prior}}\}$ and $\{\hat{z}_c^{\text{post}}, \hat{z}_{nc}^{\text{post}}\}$; then they are scaled and cascaded with the varying and invariant features $\{S_c, S_{nc}\}$ derived from the feature separator, respectively, to yield the final decoupled features $\{\text{dc}_c^{\text{prior}}, \text{dc}_{nc}^{\text{prior}}\}$ and $\{\text{dc}_c^{\text{post}}, \text{dc}_{nc}^{\text{post}}\}$. Here, we set $K = 20$. At last, the change and nonchange features are located by GT to generate decoupled feature prototypes. The prior feature prototypes are shown as follows:

$$p_c^{\text{prior}} = \frac{1}{|\Omega_c|} \sum_{i \in \Omega_c} \text{dc}_c^{\text{prior}}(i) \tag{10}$$

$$p_{nc}^{\text{prior}} = \frac{1}{|\Omega_{nc}|} \sum_{j \in \Omega_{nc}} \text{dc}_{nc}^{\text{prior}}(j) \tag{11}$$

where Ω_c and Ω_{nc} are the sets of change and nonchange features, respectively, $|\Omega_c|$ and $|\Omega_{nc}|$ represent their set sizes. $\text{dc}_c^{\text{prior}}(i) \in \mathbb{R}^{1\times c}$ indicates the ith change feature based on prior decoupling. $p_c^{\text{prior}} \in \mathbb{R}^{1\times c}$ represent the change prototypes based on prior decoupling. Similarly, the posterior decoupling feature and prototypes can be obtained.

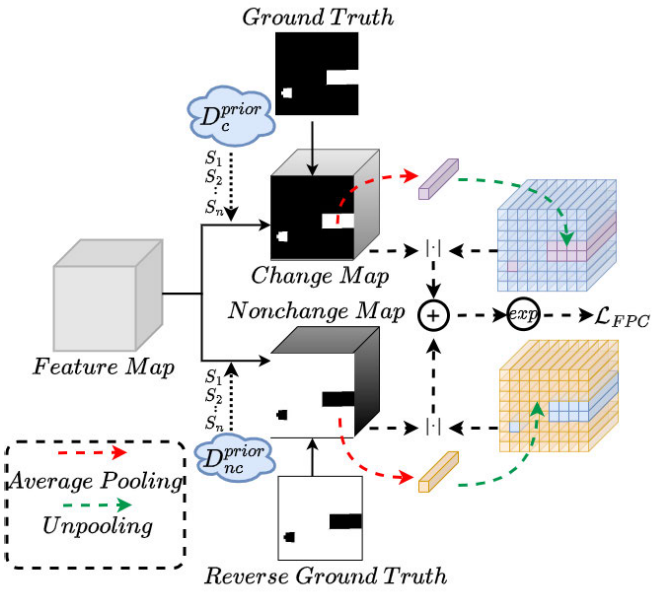


Fig. 3. Structural diagram of the prototype loss for prior decoupled features.

In the testing phase, since the change prediction is carried out by relying on the prior distribution while being separated from the posterior distribution, the prototype loss here mainly constrains prior decoupled features. The prototype constraints for changed features are as follows:

$$\mathcal{L}_{\text{FPC}}^{\text{pr}_c} = \frac{1}{N_{\text{un}}} \sum_{k \in [\text{Unchanged}]} \exp\left(\frac{\left\|\text{dc}_c^{\text{prior}}(k) - 0\right\|_2}{t}\right) + \frac{1}{N_c} \sum_{k \in [\text{Changed}]} \exp\left(\frac{\left\|\text{dc}_c^{\text{prior}}(k) - p_c^{\text{prior}}\right\|_2}{t}\right). \tag{12}$$

The prototype constraints for the invariant features based on prior features are as follows:

$$\mathcal{L}_{\text{FPC}}^{\text{pr}_{nc}} = \frac{1}{N_{\text{un}}} \sum_{k \in [\text{Unchanged}]} \exp\left(\frac{\left\|\text{dc}_{nc}^{\text{prior}}(k) - p_{nc}^{\text{prior}}\right\|_2}{t}\right) + \frac{1}{N_c} \sum_{k \in [\text{Changed}]} \exp\left(\frac{\left\|\text{dc}_{nc}^{\text{prior}}(k) - 0\right\|_2}{t}\right) \tag{13}$$

where N_c and N_{un} represent the sets of changing and invariant pixels, respectively; 0 and 1 represent the invariant and change pixels, respectively; and t is the temperature coefficient. The overall prototype constraint loss function for decoupled features is as follows:

$$\mathcal{L}_{\text{FPC}}^{\text{prior}} = \mathcal{L}_{\text{FPC}}^{\text{pr}_c} + \mathcal{L}_{\text{FPC}}^{\text{pr}_{nc}}. \tag{14}$$

D. Exceed-Expectation (EE) Loss

Most loss functions would expect the prediction probability of 1 for positive samples and 0 for negative samples. Taking cross-entropy as an instance, when samples are misclassified, the cross-entropy will calculate a relatively large loss value, which in turn helps the model adjust network

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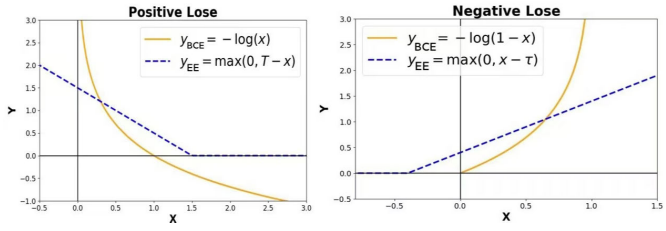


Fig. 4. Comparison Chart of $\mathcal{L}_{\mathrm{EE}}$ and $\mathcal{L}_{\mathrm{BCE}}$.

parameters. However, when samples are correctly classified or even approach the expected values, the cross-entropy will drop sharply, affecting the convergence speed and even leading to the blurred classification boundary. Motivated by this, the predicted probability targets for the positive and negative examples are pushed and pulled to T and τ instead of 1 and 0 by the proposed super-expected loss, respectively. When $T > 1$ or $\tau < 0$, it is obvious that the predicted probabilities cannot reach the corresponding values. However, a larger loss value will be calculated in each iteration, which will help improve the model's performance and accelerate convergence within a certain range. The loss function is as follows:

$$\mathcal{L}_{\mathrm{EE}} = \frac{1}{N_{\mathrm{un}}} \sum_{k \in [\mathrm{Unchanged}]} (1 - Y_k) \max(0, P_k - \tau) + \frac{1}{N_{\mathrm{c}}} \sum_{k \in [\mathrm{Changed}]} Y_k \max(0, T - P_k) \quad (15)$$

where Y_k represents the kth changing or invariant pixel in the label, and P_k represents the predicted probability for the kth pixel by the model.

The comparison between cross-entropy and EE loss is shown in Fig. 4. When there is a misdetection or omission situation, i.e., the gap between the prediction and GT is large, it is obvious that the loss value obtained from the cross-entropy is larger, which is more conducive to the model convergence and even performance enhancement. When the predictions are relatively close to matching expectation, its loss value would drop sharply. At this stage, the over-expectation loss would be promising for model tuning by increasing the prediction expectation to obtain a larger loss value. The subsequent part will conduct experimental demonstrations on the parameters τ and T.

In the testing phase, the CD prediction is mainly carried out through the prior distribution. Hence, in the training phase, we separate the prior and posterior losses to obtain the final loss function $\mathcal{L}_T$

$$\mathcal{L}_T = \lambda_1 * \left(\mathcal{L}_{\mathrm{BCE}}^{\mathrm{post}} + \mathcal{L}_{\mathrm{EE}}^{\mathrm{post}}\right) + \lambda_2 * \mathcal{L}_{\mathrm{kl}} + \lambda_3 * \mathcal{L}_{\mathrm{FPC}}^{\mathrm{prior}} + \lambda_4 * \left(\mathcal{L}_{\mathrm{BCE}}^{\mathrm{prior}} + \mathcal{L}_{\mathrm{EE}}^{\mathrm{prior}}\right). \quad (16)$$

IV. Experimental Setup

A. Dataset

Each dataset contains two images taken at different times in the same area and their corresponding CD results. The dataset details are as follows.

1) LEVIR-CD Dataset: The LEVIR-CD is a public large-scale building CD dataset. It contains 637 pairs of ultra-high resolution (0.5 m/pixel) RS images with a size of 1024 × 1024. The time span of these images is 5–14 years. These images are sourced from 20 different areas in multiple cities in Texas, USA, covering 31 333 changed building instances. It is worth noting that LEVIR-CD covers various types of buildings with significant land use changes, such as villa residences, high-rise apartments, small garages, and large warehouses. Each image pair was cropped into 16 nonoverlapping image pairs with a size of 256 × 256. And 7120/1024/2048 pairs of images were, respectively, used for training/validation/testing.

2) CDD Dataset: The CDD is a public CD dataset composed of satellite images taken during seasonal alternations, with a spatial resolution ranging from 0.03 to 1 m/pixel. The sizes of the changed areas in CDD vary, and it includes buildings, roads, vehicles, and so on. The image data was divided into 10 000/2998/3000 pairs of patches with a size of 256 × 256, which were, respectively, used for training/validation/testing.

3) SYSU Dataset: The SYSU dataset contains 20 000 pairs of aerial images with a size of 256 × 256 pixels. These images have a spatial resolution of 0.5 m and cover three bands. They were taken from 2007 to 2014 in Hong Kong. The change types are rich and diverse, such as the emergence of new urban buildings, the suburb expansion, the continuous road widening, and large-scale reclamation projects. During data processing, the images were randomly divided into training, validation, and test set according to a ratio of 6:2:2.

B. Experimental Parameters

The DSFDcd is implemented based on PyTorch and trained using a single NVIDIA RTX4090 GPU. The SGD optimizer is utilized with the momentum set to 0.9, the weight decay set to $5e^{-4}$, and the initial learning rate set to 0.0001. The learning rate of the optimizer is dynamically adjusted, which is decayed by a factor of 0.7 every 10 training epochs. In addition, the training epoch for each dataset is set to 200, and the batch size is 6.

C. Evaluation Metrics

Five evaluation metrics, namely, precision (P), recall (Re), $F1$-score, overall accuracy (OA), and intersection over union (IoU), are adopted to comprehensively evaluate the experimental results. The definitions of the above indicators are as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \quad (17)$$

$$\mathrm{Re} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \quad (18)$$

$$F1 = \frac{2 \times P \times \mathrm{Re}}{P + \mathrm{Re}} \quad (19)$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \quad (20)$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \quad (21)$$

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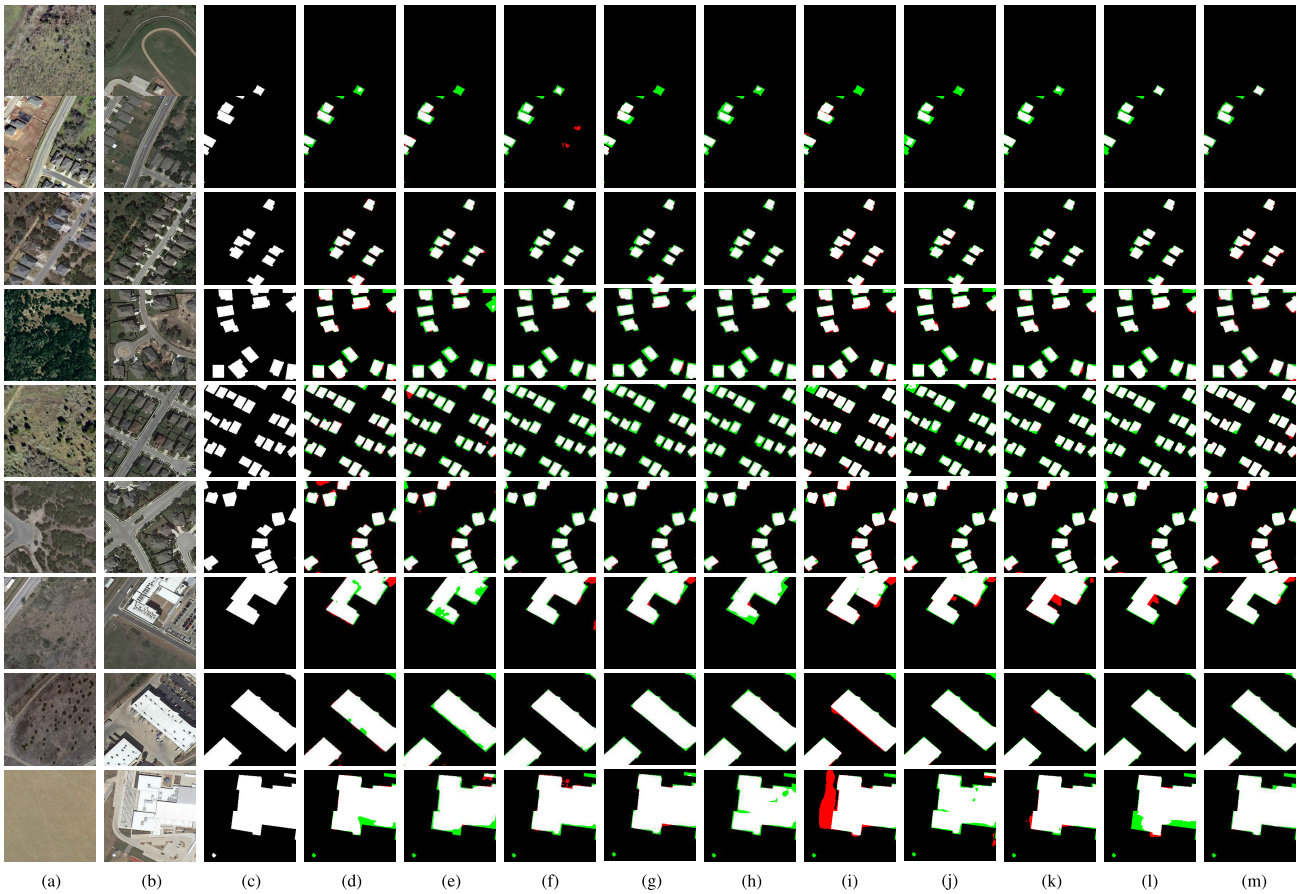


Fig. 5. Detection results on the LEVIR-CD using different methods. (a) T1 image. (b) T2 image. (c) Ground truth. (d) DTCDSCN. (e) BIT. (f) ChangeFormer. (g) ICIFNet. (h) DMINet. (i) USSFC-Net. (j) ChangeMamba. (k) Convformer-CD. (l) STRobustNet. (m) Ours.

where TP, FP, TN, and FN refer to the numbers of true positives, false positives, true negatives, and false negatives, respectively.

V. Experimental Results

A. Comparison Methods

To verify the DSFDcd effectiveness, it is compared with several state-of-the-art CD methods, including DTCDSCN [55], BIT [56], Change Former [57], ICIFNet [58], DMINet [59], USSFC-Net [60], ChangeMamba [61], Convformer-CD [62], and STRobustNet [63]. For fairness, all comparison methods use the authors' released codes and set parameters as per original literatures.

Table I shows the quantitative indicators of different models on three datasets. This model performs outstandingly in comprehensive indicators such as $F1$, OA, and IoU across multiple datasets. For example, on the LEVIR dataset, the $F1$ reaches up to 89.11. This is attributed to mechanisms such as feature decoupling, which help to improve the problem of blurred semantic boundaries, enabling the model to accurately locate the changed areas.

The intuitive comparison of different methods on various datasets is shown in Figs. 5–7. On the LEVIR-CD dataset, BIT, ICIFNet, DMINet, and ChangeMamba have severe false negatives in small target areas, BIT, ChangeFormer, ICIFNet, USSFC-Net, ChangeMamba, CovFormer-CD, and STRobustNet show varying degrees of false positives in larger target areas, while USSFC-Net has high false positives in large target areas. In addition, near the edges of densely arranged complex buildings, BIT is prone to false positives and DMINet to false negatives. When there is light interference, DTCDSCN has obvious false positives; BIT, DMINet, and USSFC-Net have obvious false negatives. Fortunately, the DSFDcd results show clear boundaries and the low false positive and negative rate. The CDD dataset highlights the impact of seasonal changes. BIT, ChangeFormer, and ICIFNet can identify significant change areas but overlook smaller ones. DMINet has made progress but is insufficient in detecting subtle changes. In larger change areas, ChangeFormer, ICIFNet, and USSFC-Net have issues like false positives of edge targets and loss of unchanged objects, while DSFDcd shows excellent detection performance in both large and small change areas and strong resistance to seasonal changes. On the SYSU dataset, DTCDSCN and ICIFNet have serious false positives under illumination disturbance, and false negatives are likely to occur when detecting large-scale target changes in complex scenes. The DSFDcd performs better in terms of target integrity and boundary accuracy.

Table II reports the running efficiency of different models, including parameter number, floating-point computation, and running time (with respect to the LEVIR dataset). It can be noticed that the parameter number of our model is moderate, but the FLOPs are not dominant. This indicates that the model is not very demanding in terms of GPU memory during

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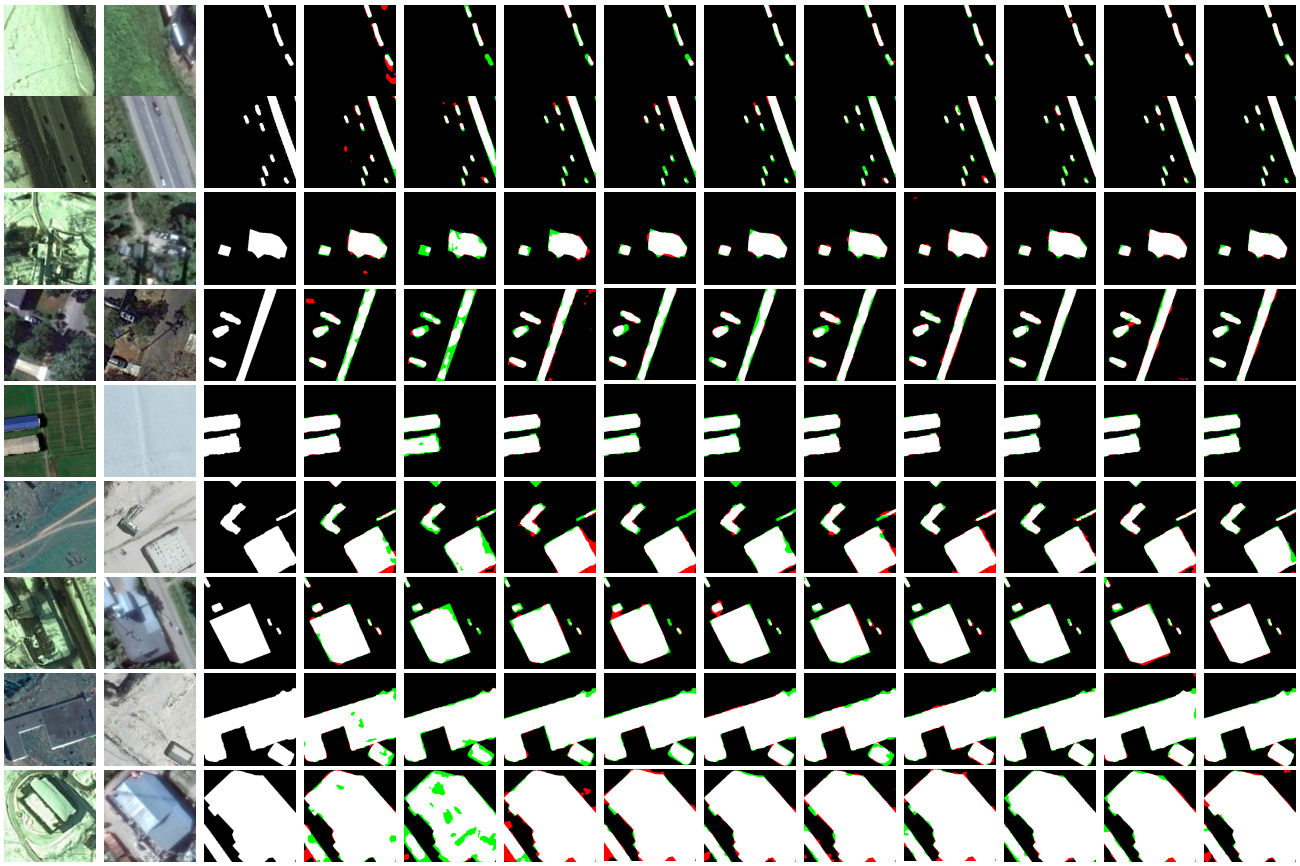


Fig. 6. Detection results on the CDD using different methods. (a) T1 image. (b) T2 image. (c) Ground truth. (d) DTCDSCN. (e) BIT. (f) ChangeFormer. (g) ICIFNet. (h) DMINet. (i) USSFC-Net. (j) ChangeMamba. (k) Convformer-CD. (l) STRobustNet. (m) Ours.

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON VARIOUS DATASETS

Methods	LEVIR - CD					CDD					SYSU				
	P.	Re.	F1	OA	IoU	P.	Re.	F1	OA	IoU	P.	Re.	F1	OA	IoU
DTCDSCN [55]	72.64	**86.66**	79.03	98.21	65.25	92.41	91.75	92.08	98.13	85.51	83.19	77.25	80.11	90.96	66.81
BIT [56]	79.75	75.51	77.57	98.20	63.30	95.17	83.99	89.23	97.61	80.55	77.69	74.53	76.08	88.95	61.40
Change Former [57]	78.70	80.21	79.50	98.38	65.87	92.51	89.60	91.03	97.92	83.54	81.31	70.40	75.46	89.20	60.60
ICIFNet [58]	76.65	81.25	78.88	98.30	65.17	94.71	88.56	91.53	98.07	84.49	84.60	70.70	77.03	90.06	62.64
DMINet [59]	79.23	77.61	78.41	98.33	64.53	95.22	90.41	92.75	98.33	86.49	86.06	71.13	77.88	90.47	63.89
USSFC-Net [60]	90.13	84.19	87.06	98.73	77.13	95.49	91.29	93.34	98.40	87.53	78.46	77.25	77.85	89.63	63.73
ChangeMamba [61]	75.99	83.12	79.39	98.32	65.83	92.50	87.83	90.10	97.72	81.99	83.23	76.63	79.80	90.85	66.39
Convformer-CD [62]	91.96	85.42	88.57	98.88	79.66	94.81	93.14	93.96	**98.59**	88.63	**86.22**	73.49	79.35	90.98	65.76
STRobustNet [63]	**93.15**	83.91	88.29	98.87	79.05	93.32	**94.01**	93.66	98.50	88.08	82.41	74.42	78.21	90.22	64.22
Ours	92.50	85.95	**89.11**	**98.93**	**80.34**	**96.69**	91.57	**94.06**	98.57	**88.81**	82.43	**78.57**	**80.46**	**91.00**	**67.31**

*All values are reported in percentage (%). Color convention: **best**, 2-nd best.

training, but requires more floating-point computations during forward propagation. In addition, the running time is in the acceptable range. In conclusion, how to optimize the code to improve FLOPs will be one of the key research efforts. A more straightforward comparison is shown in Fig. 8, where the larger bubbles indicate the higher $F1$ value. It is not difficult to see that the performance of our model is superior, and generating features through distribution sampling keeps both the model parameters and FLOPs within an acceptable range.

B. Ablation Experiments

In this article, the module effectiveness (i.e., decoupled feature distribution, prototype loss, and EE loss) is verified through ablation experiments. As shown in Table III, the symbol "✓" indicates that the corresponding module is adopted. The details of the ablation experiments are explained as follows: if there is no stochastic sampling from the decoupled distribution to extract features, the feature de-entanglement is only implemented based on the convolutional feature separator; if either $\mathcal{L}_{\text{FPC}}$ or $\mathcal{L}_{\text{EE}}$ is not included, they will be removed directly. Notably, $\mathcal{L}_{\text{BCE}}$ is always present to guarantee the basic experimental effect during the ablation experiments of the loss function. The experimental results show that the performance of the model containing only a single module is generally worse than the one including two modules; while the outcome outperforms the others when all modules of the model are

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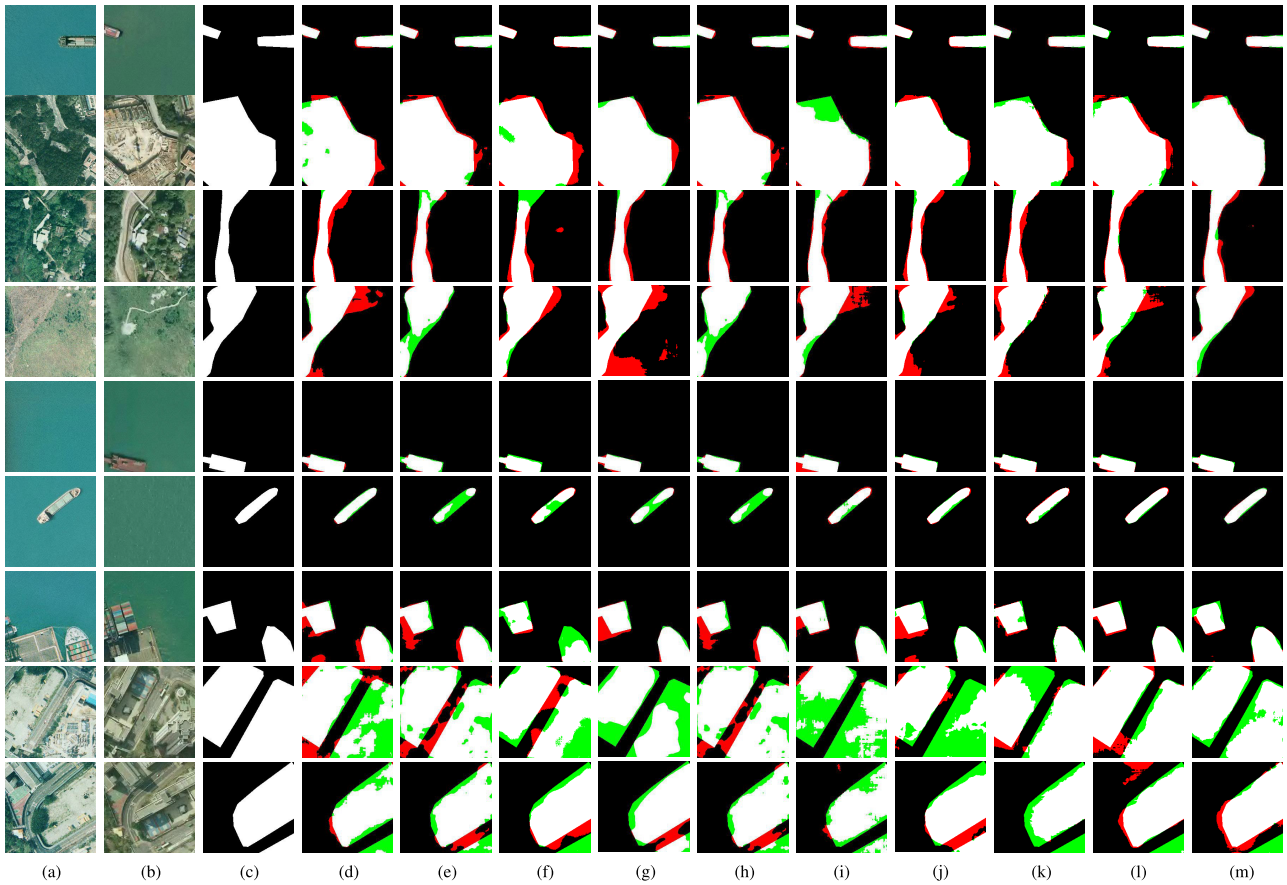


Fig. 7. Detection results on the SYSUCD using different methods. (a) T1 image. (b) T2 image. (c) Ground truth. (d) DTCDSCN. (e) BIT. (f) ChangeFormer. (g) ICIFNet. (h) DMINet. (i) USSFC-Net. (j) ChangeMamba. (k) Convformer-CD. (l) STRobustNet. (m) Ours.

TABLE II

COMPARATIVE EXPERIMENTS ON THE COMPUTATIONAL EFFICIENCY OF DIFFERENT MODELS

Methods	FLOPs(G)	Params(M)	Runtime(min/epoch)
DTCDSCN	20.44	41.07	1.00
BIT	15.10	17.99	1.10
Change Former	21.18	29.75	3.90
ICIFNet	25.36	23.82	2.60
DMINet	14.55	6.24	1.20
USSFC-Net	4.86	1.52	1.90
ChangeMamba	45.74	17.13	3.10
Convformer-CD	5.14	37.22	2.00
STRobustNet	13.19	5.23	1.50
Ours	29.96	8.94	2.80

TABLE III

ABLATION EXPERIMENTS OF EACH MODULE ABOUT DSFDCD

Model DFD	FPC	EE		LEVIR-CD F1 / IoU	CDD F1 / IoU	SYSU F1 / IoU
✓	/	/		75.21/60.25	89.90/81.66	79.32/65.56
/	✓	/		74.59/59.57	78.17/64.17	67.28/50.70
/	/	/	✓	70.98/55.29	68.50/52.10	64.06/47.13
/	✓	/	✓	84.67/73.43	91.91/85.04	79.34/65.76
✓	/	/	✓	85.84/75.19	92.25/85.62	79.56/66.08
✓	/	✓	/	87.45/77.69	92.59/86.39	80.19/66.95
✓	/	✓	/	✓	**89.11/80.34**	**94.06/88.81**	**80.46/67.31**

available. In short, each module makes its own individual contribution to improving model performance regardless of the dataset.

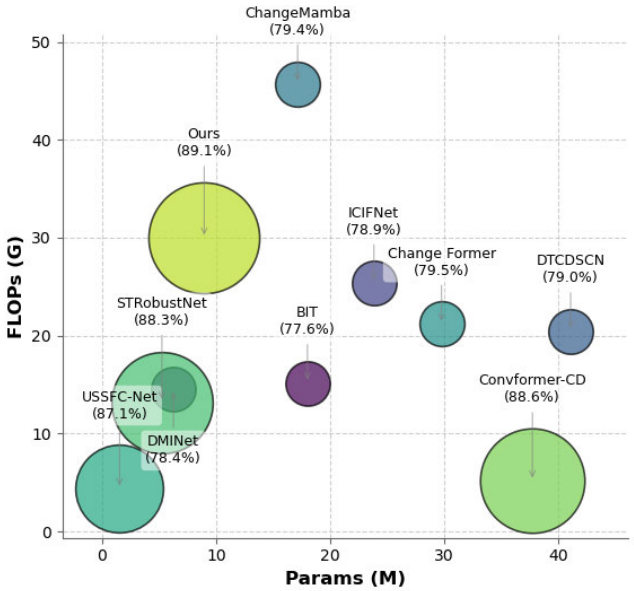


Fig. 8. Visual comparison of the model efficiency and the comprehensive indicator $F1$.

C. Parameter Analysis

Table IV shows the impact of the EE loss with different parameters τ and T on the model evaluation about LEVIR-CD. It should be noted that the ratio of BCE to hyper-expectation loss is set to 15:1. While keeping $\tau = 0$, we tested the influence of different values of T to explore the maximum

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TABLE IV

IMPACTS OF DIFFERENT HYPERPARAMETERS T AND τ ON THE MODEL WHEN USING THE EE LOSS FUNCTION ON THE LEVIR-CD DATASET

T	τ	P.	Re.	F1	IoU
0.5	0	**95.51**	77.54	85.59	74.81
1	0	91.55	**87.20**	89.32	80.70
2	0	92.82	86.17	**89.37**	**80.76**
3.5	0	92.50	85.95	89.11	80.34
2	-2	**93.23**	83.97	88.36	79.33
2	-1	91.39	**87.58**	**89.45**	**80.89**

TABLE V

INFLUENCE OF REGULARIZATION COEFFICIENT ON QUANTITATIVE EXPERIMENTAL RESULTS REGARDING DIFFERENT DATASETS

Dataset	λ_1	λ_2	λ_3	λ_4	P.	Re.	F1	IoU
LEVIR	1	1	1	1	91.85	85.26	88.43	79.26
	3	2	2	3	**94.73**	81.69	87.73	80.15
	3	2	2	2	93.45	84.01	88.48	79.36
	2	2	2	3	92.50	**85.95**	**89.11**	**80.34**
	1	2	2	1	92.39	85.14	88.62	79.56
	1	2	2	2	92.22	84.77	88.34	79.28
CDD	1	1	1	1	95.49	90.47	92.91	86.76
	3	2	2	3	96.69	**91.57**	**94.06**	**88.81**
	3	2	2	2	**97.23**	90.87	93.95	88.76
	2	2	2	3	96.66	91.26	93.88	88.47
	1	2	2	1	96.66	89.09	92.72	86.44
	1	2	2	2	95.88	90.42	93.07	87.22
SYSU	1	1	1	1	82.72	78.25	80.43	67.26
	3	2	2	3	82.43	78.57	**80.46**	**67.31**
	3	2	2	2	**84.32**	73.39	78.48	64.44
	2	2	2	3	80.76	**79.76**	80.26	67.02
	1	2	2	1	83.21	76.82	79.89	66.51
	1	2	2	2	82.85	77.37	80.02	66.84

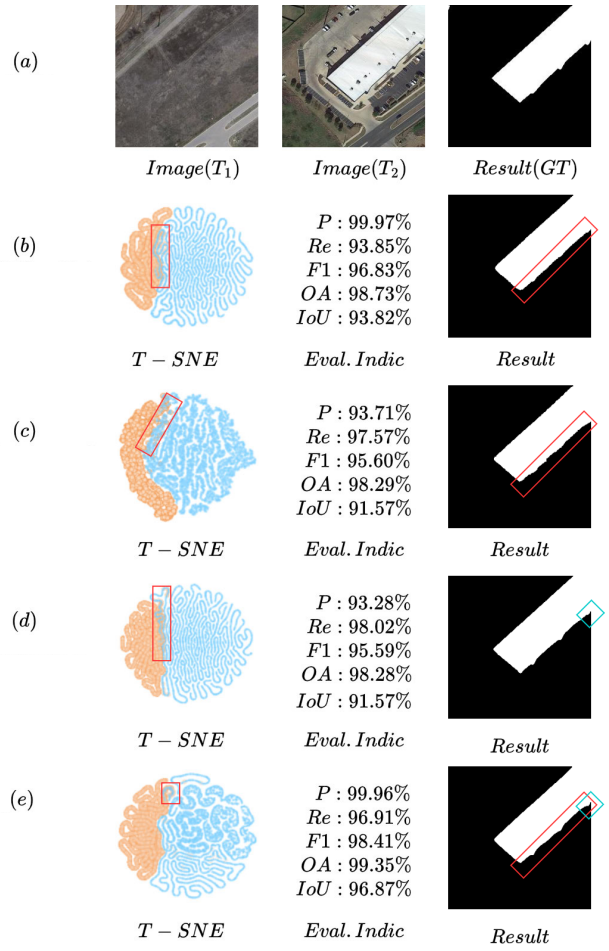


Fig. 9. Some visualization examples produced by typical deep models. (a) Bi-temporal RS images and GT. (b) Results of the USSFC-Net. (c) Results of the Convformer-CD. (d) Results of the STRobustNet. (e) Results of our DSFDcd.

target threshold, which the positive samples can be pushed and pulled. It is finally determined that comprehensive performance reaches a relatively optimal solution when $T = 2$. Similarly, another optimal value $\tau = -1$ can be obtained. The experiments show that excessive pushing and pulling may lead to an overly large loss value, which is not conducive to the stability and optimization of parameters.

D. Regularization Term Coefficients

In addition, experimental tuning for the regularization term of the loss function was carried out on three datasets, which results are shown in Table V. With these optimal comprehensive performance indicators, the parameter configuration shows that different datasets have slightly different dependencies on the posterior distribution.

E. Visualization

To reveal the CD effects and its factors more intuitively, the T-SNE dimensionality reduction technique is used for visual interpretation. The T-SNE visualization projects the high-dimensional embedded features onto the 2-D plane, which is more accessible to the researcher. As shown in Fig. 9, the yellow and blue dots on the T-SNE plane represent varying and nonvarying pixel features, respectively. Obviously, other models confuse the two types of pixels more than our model, which tends to cause blurring of semantic boundaries, thus affecting the CD performance. In terms of objective evaluation of this example, DSFDcd is ahead in the comprehensive metrics (i.e., $F1$, OA, and IOU), which is in conformity with the T-SNE visualization results. In regards to the prediction results, the difference is mainly reflected in the classification clarity of "jagged object" as marked in the red box; the prediction results of STRobustNet are also clearly bounded, although it misses a piece as marked in the blue box. In conclusion, DSFDcd separates the changing and unchanging pixels through feature decoupling, which not only makes the semantic boundary clearer but also allows the CD to be more easily interpreted.

To gain a more intuitive understanding, we selected a representative set of samples from the LEVIR-CD to visualize features generated by DSFDcd as shown in Fig. 10. The changed features s_c and unchanged features s_{nc} gained by the feature decoupler [see Fig. 10(d) and (e)] indicate that our model effectively focuses the attention of the decoupled features on the areas of interest and suppresses the interference

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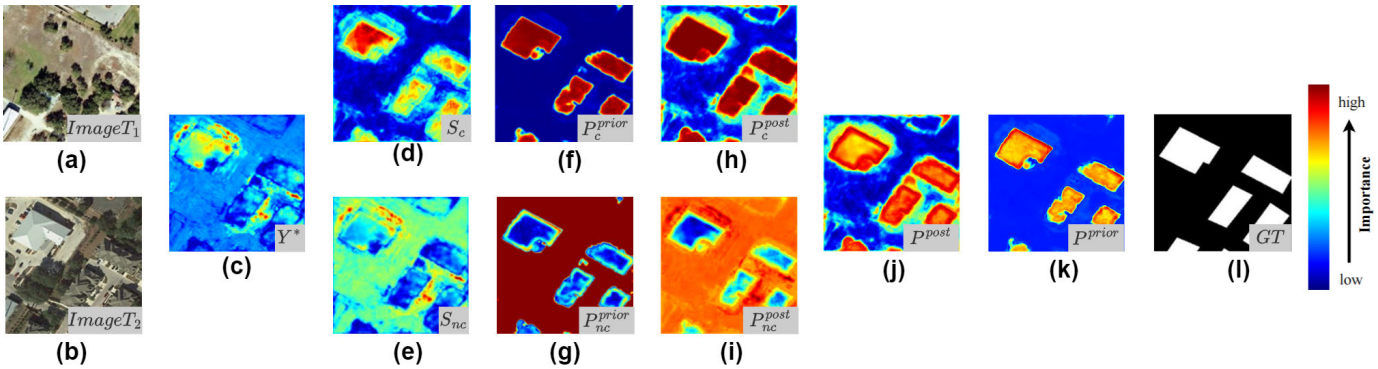


Fig. 10. Visualization of the key modules from DSFDcd. (a) and (b) Bi-temporal images; (c) differential features; (d) and (e) decoupled features from feature separator; (f) and (g) prior decoupled features under prototype constraints; (h) and (i) posterior decoupled features; (j) CD result from posterior distribution; (k) CD result from prior distribution; and (l) GT.

from the background that is irrelevant to the CD task. From Fig. 10(d)–(i), the posterior features have superior attentional focus than the separation ones, but the a priori decoupling features under prototype constraints are more fine-grained than the previous pair. The generated CD probability map in Fig. 10(j) shows more accurate target localization and finer boundaries compared to the others, indicating that DSFDcd achieves feature decoupling, combining feature extraction with distribution sampling and effectively captures the changed targets.

Although the model has achieved certain effects, there is still room for further improvement: first, a lightweight model is more conducive to practical applications; second, the human brain is the most robust and reliable system, so improving the model further according to the brain-like structure may be a better way to deal with the interference caused by problems such as illumination differences in the dual-temporal images.

VI. Conclusion

Mixed-feature decision making is likely to lead to blurred semantic boundaries and inaccurate classification. To address this issue, this article proposes an RSCD framework based on feature decoupling. First, the posterior features of the changed and unchanged regions are utilized to learn the prior probability distribution. Then, the feature separator is trained through posterior distribution resampling and prototype learning. Finally, the relevant parameters for feature extraction and decoupling under the CD task are trained based on a hybrid loss function combining BCE and EE loss. During the testing phase, the model mainly uses the feature extractor, separator, decoupling prior distribution, and CD head to complete RSCD. In addition, comparative experiments on multiple datasets show that DSFDcd performs well in both comprehensive indicators (e.g., $F1$ and IoU) and qualitative comparisons. This proves that DSFDcd is highly adaptable to detect changes about different types of targets.

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